Database Administration and Management Lab Project

Blood Bank Management System

Project Advisor

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1. Introduction:

Database Administration and Management focuses on maintaining, securing, and optimizing databases to ensure efficient and reliable data handling. In healthcare systems, especially blood banks, databases must be carefully administered to guarantee data accuracy, integrity, and availability at all times.

Blood banks manage sensitive and critical information related to donors, recipients, hospitals, and blood inventory. Any error in data handling may result in serious consequences, including delays in treatment or misuse of blood units. Therefore, a well-administered database system is essential.

This project implements a Blood Bank Management System using Microsoft SQL Server Management Studio (SSMS) with a strong focus on database administration concepts such as stored procedures, triggers, views, indexes, and joins. These features help automate processes, improve performance, and maintain consistency within the database.

2. Problem Statement:

Despite advancements in healthcare technology, many blood banks still face administrative and data management challenges. Common issues include:

- Redundant and inconsistent data due to lack of normalization
- Manual monitoring of blood availability and expiry dates
- No automated response to blood usage or request completion
- Slow query performance due to absence of indexing
- Difficulty in generating accurate reports for management
- No centralized system for easy data retrieval.

Without proper database administration techniques, even a well-designed database can perform poorly. Hence, an administrative-level solution is required to ensure smooth operation, automation, and optimization of the blood bank database.

3. Objectives:

The main objectives of this project are:

- To implement a database system using advanced **Database Administration & Management** concepts
- To use **stored procedures** for reusable and secure data operations
- To implement **triggers** for automatic updates and enforcement of business rules
- To create **views** for simplified data access and reporting
- To apply **indexes** for performance optimization

- To use **joins** for generating meaningful multi-table reports
- To gain practical experience in SQL Server database administration using a real-world case study

4. Tools & Technologies:

The tools and technologies used in this project are listed below:

Technology	Description
Database Server	Microsoft SQL Server
Management Tool	SQL Server Management Studio (SSMS)
Query Language	T-SQL (Transact-SQL)
Operating System	Windows OS

Microsoft SQL Server provides powerful administrative features such as triggers, procedures, indexing, and views, while SSMS allows easy database management, monitoring, and debugging.

5. Project Scope:

The scope of the Blood Bank Management System includes the following modules with administrative enhancements:

- **Donor Management:** Controlled insertion and validation of donor records
- **Blood Inventory Management:** Automatic updates of availability and expiry status
- **Recipient Management:** Structured handling of recipient data
- **Hospital Request Management:** Automated processing of blood requests
- **Quality Control Module:** Tracking contamination and test results
- **Administration Features:**
    - Triggers for automation
    - Stored procedures for controlled access
    - Views for reporting
    - Indexes for query optimization

6. Timeline:

The project was completed according to the following timeline:

- Requirement analysis and administrative planning
- Database schema design and normalization
- Table creation with constraints
- Sample data insertion
- Implementation of triggers, procedures, views, and indexes
- Testing and debugging queries
- Report preparation and final submission

7. Expected Deliverables:

Upon completion, the following deliverables are submitted:

- Fully implemented SQL Server database
- Tables with relationships and constraints
- Stored procedures for data operations
- Triggers for automated actions
- Views for reporting and data abstraction
- Optimized queries using indexes and joins
- Final project report and database demonstration

8. Output:

SQL Server Code:

SQLQuery1.sql - IP...oodBankDB (sa (53))*

```sql
/*************************************************************
   BLOOD BANK MANAGEMENT SYSTEM
   DATABASE ADMINISTRATION & MANAGEMENT LAB
*************************************************************/

-------------------------------------------------------------
-- DATABASE
-------------------------------------------------------------
IF DB_ID('BloodBankDB') IS NULL
    CREATE DATABASE BloodBankDB;
GO
USE BloodBankDB;
GO

-------------------------------------------------------------
-- DROP TABLES (RESET)
-------------------------------------------------------------
IF OBJECT_ID('QualityControl','U') IS NOT NULL DROP TABLE QualityControl;
IF OBJECT_ID('HospitalRequest','U') IS NOT NULL DROP TABLE HospitalRequest;
IF OBJECT_ID('BloodInventory','U') IS NOT NULL DROP TABLE BloodInventory;
IF OBJECT_ID('Recipient','U') IS NOT NULL DROP TABLE Recipient;
IF OBJECT_ID('Hospital','U') IS NOT NULL DROP TABLE Hospital;
IF OBJECT_ID('Donor','U') IS NOT NULL DROP TABLE Donor;
GO

-------------------------------------------------------------
-- TABLES
-------------------------------------------------------------
CREATE TABLE Donor (
    DonorID INT IDENTITY PRIMARY KEY,
    FirstName VARCHAR(50),
```

SQLQuery1.sql - IP...oodBankDB (sa (53))*

```sql
-------------------------------------------------------------
CREATE TABLE Donor (
    DonorID INT IDENTITY PRIMARY KEY,
    FirstName VARCHAR(50),
    LastName VARCHAR(50),
    Gender CHAR(1),
    DateOfBirth DATE,
    BloodGroup VARCHAR(5),
    DiseaseStatus VARCHAR(50),
    ContactNumber VARCHAR(20),
    Email VARCHAR(100),
    Address VARCHAR(255),
    RegistrationDate DATE DEFAULT GETDATE()
);

CREATE TABLE Hospital (
    HospitalID INT IDENTITY PRIMARY KEY,
    HospitalName VARCHAR(100),
    ContactNumber VARCHAR(20),
    Address VARCHAR(255)
);

CREATE TABLE Recipient (
    RecipientID INT IDENTITY PRIMARY KEY,
    FirstName VARCHAR(50),
    LastName VARCHAR(50),
    Gender CHAR(1),
    DateOfBirth DATE,
    BloodGroup VARCHAR(5),
    ContactNumber VARCHAR(20),
    Email VARCHAR(100),
```

SQLQuery1.sql - IP...oodBankDB (sa (53))*

```sql
-- INSERT DATA
-------------------------------------------------------------
INSERT INTO Hospital VALUES
('City Hospital','0421111111','Lahore'),
('Care Medical Center','0422222222','Lahore'),
('Shifa International','0213333333','Karachi'),
('Faisal Hospital','0414444444','Faisalabad'),
('Medicare Hospital','0515555555','Islamabad'),
('Life Care Hospital','0616666666','Multan'),
('National Hospital','0427777777','Lahore'),
('Al Noor Hospital','0218888888','Karachi');

INSERT INTO Donor VALUES
('Ali','Khan','M','1995-05-10','A+','None','03001234567','ali@mail.com','Lahore',GETDATE()),
('Sara','Ahmed','F','1998-07-15','O-','None','03111234567','sara@mail.com','Lahore',GETDATE()),
('Umer','Rashid','M','1988-02-20','B+','Diabetes','03211234567','umer@mail.com','Lahore',GETDATE()),
('Fatima','Malik','F','1999-11-11','AB+','None','03451234567','fatima@mail.com','Karachi',GETDATE()),
('Ahmed','Zubair','M','2000-12-25','O+','Hepatitis','03561234567','ahmed@mail.com','Karachi',GETDATE()),
('Ayesha','Malik','F','1999-04-18','B+','None','03021234567','ayesha@mail.com','Islamabad',GETDATE()),
('Bilal','Hassan','M','1996-08-09','A-','None','03331234567','bilal@mail.com','Faisalabad',GETDATE()),
('Hina','Yousaf','F','2001-01-01','O+','None','03411234567','hina@mail.com','Multan',GETDATE());

INSERT INTO BloodInventory VALUES
(1,'2025-06-01','2025-07-01','No','Available'),
(2,'2025-06-05','2025-07-05','No','Available'),
(3,'2025-05-20','2025-06-20','Yes','Used'),
(4,'2025-06-10','2025-07-10','No','Available'),
(5,'2025-05-15','2025-06-15','Yes','Used'),
(6,'2025-06-12','2025-07-12','No','Available'),
(7,'2025-06-14','2025-07-14','No','Available'),
(8,'2025-06-16','2025-07-16','No','Available');
```

SQLQuery1.sql - IP...oodBankDB (sa (53))*

```sql
-- INDEXES
-------------------------------------------------------------
CREATE INDEX idx_Donor_BloodGroup ON Donor(BloodGroup);
CREATE INDEX idx_Blood_Availability ON BloodInventory(AvailabilityStatus);
GO

-------------------------------------------------------------
-- VIEWS
-------------------------------------------------------------
IF OBJECT_ID('vw_AvailableBlood','V') IS NOT NULL DROP VIEW vw_AvailableBlood;
GO
CREATE VIEW vw_AvailableBlood AS
SELECT b.BloodID, d.BloodGroup, b.AvailabilityStatus
FROM BloodInventory b
JOIN Donor d ON b.DonorID = d.DonorID
WHERE b.AvailabilityStatus='Available';
GO

IF OBJECT_ID('vw_HealthyDonors','V') IS NOT NULL DROP VIEW vw_HealthyDonors;
GO
CREATE VIEW vw_HealthyDonors AS
SELECT * FROM Donor WHERE DiseaseStatus='None';
GO

-------------------------------------------------------------
-- STORED PROCEDURES
-------------------------------------------------------------
IF OBJECT_ID('GetBloodByGroup','P') IS NOT NULL
    DROP PROCEDURE GetBloodByGroup;
GO
CREATE PROCEDURE GetBloodByGroup
```

SQLQuery1.sql - IP...oodBankDB (sa (53))*

```sql
-- TRIGGER
-------------------------------------------------------------
IF OBJECT_ID('trg_UpdateAvailability','TR') IS NOT NULL
    DROP TRIGGER trg_UpdateAvailability;
GO
CREATE TRIGGER trg_UpdateAvailability
ON HospitalRequest
AFTER INSERT
AS
BEGIN
    UPDATE BloodInventory
    SET AvailabilityStatus='Used'
    WHERE BloodID IN (SELECT BloodID FROM inserted);
END;
GO

-------------------------------------------------------------
-- COMPLEX JOIN REPORT
-------------------------------------------------------------
SELECT
    d.FirstName + ' ' + d.LastName AS DonorName,
    d.BloodGroup,
    h.HospitalName,
    r.FirstName + ' ' + r.LastName AS RecipientName,
    hr.Status
FROM HospitalRequest hr
JOIN Hospital h ON hr.HospitalID = h.HospitalID
JOIN Recipient r ON hr.RecipientID = r.RecipientID
JOIN BloodInventory b ON hr.BloodID = b.BloodID
JOIN Donor d ON b.DonorID = d.DonorID;
GO
```

Tables Output:

	DonorName	BloodGroup	HospitalName	RecipientName	Status
1	Ali Khan	A+	City Hospital	Ahmad Raza	Pending
2	Sara Ahmed	O-	Care Medical Center	Zainab Ali	Completed
3	Fatima Malik	AB+	Shifa International	Usman Iqbal	Pending
4	Ayesha Malik	B+	Faisal Hospital	Maryam Khalid	Pending
5	Bilal Hassan	A-	Medicare Hospital	Hamza Javed	Completed

	DonorID	FirstName	LastName	Gender	DateOfBirth	BloodGroup	DiseaseStatus	ContactNumber	Email	Address	RegistrationDate
1	1	Ali	Khan	M	1995-05-10	A+	None	03001234567	ali@mail.com	Lahore	2026-01-31
2	2	Sara	Ahmed	F	1998-07-15	O-	None	03111234567	sara@mail.com	Lahore	2026-01-31
3	3	Umer	Rashid	M	1988-02-20	B+	Diabetes	03211234567	umer@mail.com	Lahore	2026-01-31
4	4	Fatima	Malik	F	1999-11-11	AB+	None	03451234567	fatima@mail.c...	Karachi	2026-01-31
5	5	Ahmed	Zubair	M	2000-12-25	O+	Hepatitis	03561234567	ahmed@mail....	Karachi	2026-01-31
6	6	Ayesha	Malik	F	1999-04-18	B+	None	03021234567	ayesha@mail....	Islama...	2026-01-31
7	7	Bilal	Hassan	M	1996-08-09	A-	None	03331234567	bilal@mail.com	Faisal...	2026-01-31
8	8	Hina	Yousaf	F	2001-01-01	O+	None	03411234567	hina@mail.com	Multan	2026-01-31

	RecipientID	FirstName	LastName	Gender	DateOfBirth	BloodGroup	ContactNumber	Email	Address	HospitalID
1	1	Ahmad	Raza	M	2000-02-20	A+	03011223344	ahmad@mail.com	Lahore	1
2	2	Zainab	Ali	F	1997-03-12	O-	03011223355	zainab@mail.com	Lahore	2
3	3	Usman	Iqbal	M	2002-03-15	B+	03019876543	usman@mail.com	Karachi	3
4	4	Maryam	Khalid	F	2001-09-10	AB+	03014567890	maryam@mail.c...	Islama...	5
5	5	Hamza	Javed	M	1998-04-22	O+	03114567890	hamza@mail.com	Multan	6

	HospitalID	HospitalName	ContactNumber	Address
1	1	City Hospital	0421111111	Lahore
2	2	Care Medical Center	0422222222	Lahore
3	3	Shifa International	0213333333	Karachi
4	4	Faisal Hospital	0414444444	Faisal...
5	5	Medicare Hospital	0515555555	Islama...
6	6	Life Care Hospital	0616666666	Multan
7	7	National Hospital	0427777777	Lahore
8	8	Al Noor Hospital	0218888888	Karachi

	BloodID	DonorID	CollectionDate	ExpiryDate	ContaminationStatus	AvailabilityStatus
1	1	1	2025-06-01	2025-07-01	No	Available
2	2	2	2025-06-05	2025-07-05	No	Available
3	3	3	2025-05-20	2025-06-20	Yes	Used
4	4	4	2025-06-10	2025-07-10	No	Available
5	5	5	2025-05-15	2025-06-15	Yes	Used
6	6	6	2025-06-12	2025-07-12	No	Available
7	7	7	2025-06-14	2025-07-14	No	Available
8	8	8	2025-06-16	2025-07-16	No	Available

	QCID	BloodID	TestDate	TestResult	Remarks
1	1	1	2025-06-02	Passed	Safe
2	2	3	2025-05-21	Failed	Contaminated
3	3	4	2025-06-11	Passed	Safe
4	4	6	2025-06-13	Passed	Safe
5	5	8	2025-06-17	Passed	Safe

	BloodID	BloodGroup	AvailabilityStatus
1	1	A+	Available
2	2	O-	Available
3	4	AB+	Available
4	6	B+	Available
5	7	A-	Available
6	8	O+	Available

	DonorID	FirstName	LastName	Gender	DateOfBirth	BloodGroup	DiseaseStatus	ContactNumber	Email	Address	RegistrationDate
1	1	Ali	Khan	M	1995-05-10	A+	None	03001234567	ali@mail.com	Lahore	2026-01-31
2	2	Sara	Ahmed	F	1998-07-15	O-	None	03111234567	sara@mail.com	Lahore	2026-01-31
3	4	Fatima	Malik	F	1999-11-11	AB+	None	03451234567	fatima@mail....	Karachi	2026-01-31
4	6	Ayesha	Malik	F	1999-04-18	B+	None	03021234567	ayesha@mail...	Islama...	2026-01-31
5	7	Bilal	Hassan	M	1996-08-09	A-	None	03331234567	bilal@mail.com	Faisal...	2026-01-31
6	8	Hina	Yousaf	F	2001-01-01	O+	None	03411234567	hina@mail.com	Multan	2026-01-31

9. Conclusion

The Blood Bank Management System developed for the Database Administration & Management Lab successfully demonstrates the application of advanced SQL Server features in a real-world healthcare scenario. By implementing triggers, stored procedures, views, indexes, and joins, the system ensures automation, performance optimization, and data integrity.

This project enhanced our understanding of database administration responsibilities such as maintaining consistency, improving efficiency, and supporting reliable data access. The use of SQL Server Management Studio provided hands-on experience in managing and optimizing relational databases.

Overall, the project highlights the importance of database administration in mission-critical systems like blood banks, where timely and accurate data management can directly impact human lives.